

Module 1

Introduction Database Concepts

1.1 Introduction, Characteristics of databases, File system v/s Database system, Data abstraction and data Independence, DBMS system architecture, Database Administrator

Introduction

A database is an organized collection of data that can be modified, retrieved, or updated. Data, DBMS, and applications associated with them together form the database concept. The data, stored in the database, is in the row and column format, which is called a table. Every website, which needs us to sign up, uses a database. There is no internet without databases.

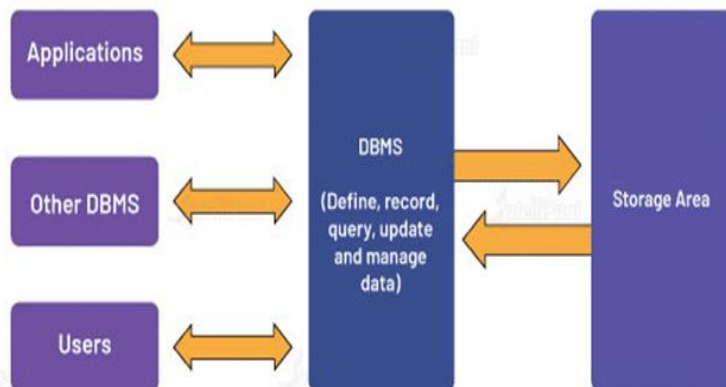
For instance, a college will have to keep the information about its students, including roll number, name, age, blood group, etc. The college will also need to keep the details of the professors and infrastructure. The details, that the college has, can be stored in a database named College, or if it is just the student details, then it can be named Students. All such details should be in a structured format, such as tables, in a hierarchy.

What is a Database Management System?

Database Management System (DBMS) is a software application that is used to create, access, maintain, and manage databases. We can install it like any other application and use [SQL](#) for all the operations that we intend to perform on databases. DBMS accepts the incoming data either from an application or from a user who is manually entering it.

- It does not matter if it is a large or small DBMS, with its help, we can store and retrieve data and make changes whenever we want to.
- Some commands are predefined in DBMS, and these commands can be used to manipulate the database. These commands are also the interface between the database and end-users to establish communication.

Database Management System



Components of DBMS

There are five major components of DBMS:

- **Hardware** – Hardware includes the actual physical computer system that is used to access the database. Mainframes to microcomputers are utilized in DBMS. Oftentimes large storage devices are used to store a huge amount of data.
- **Software** – The DBMS is the Software here. It helps connect the physical database to the end users. Whenever a user wants to access the database, access has to be granted by the software.
- **Procedures** – All the instructions that are given to access the DBMS, fall under Procedures. From installing DBMS to generating reports, all of these instructions are a part of the procedures involved in DBMS.
- **Data** – Data is the information that is managed by DBMS. DBMS helps in managing data and easing access to useful information. Metadata, the information stored in DBMS for better comprehension of the data, is also stored in DBMS itself.
- **Database Access Language** – The language that is used to write commands on the DBMS to access, utilize, manage, update and delete data, is called the Database Access Language. Commands are written using the Database Access Language and submitted to the DBMS so they can be executed by the DBMS.

Types of DBMS

There are broadly four categories or types of DBMS:

1. Hierarchical databases
2. Network databases
3. Relational databases ([RDBMS](#))
4. Object-oriented databases

- **Hierarchical databases:** This type of DBMS showcases a parent-child type of relationship. This relationship forms a tree-like structure where the nodes (leaves) of the tree represent records and the fields are represented by branches.
- **Network databases:** This style of DBMS embraces several partnerships where it is possible to connect multiple user records at the same time in parallel.
- **Relational databases (RDBMS):** This type of DBMS helps users to locate and manipulate data that has connections with another piece of data in the database. It uses tables for storing the data in a row-and-column format.
- **Object-oriented databases:** In this type of DBMS, data is stored in individual components called objects, where each object is a piece of data with some instructions for the tasks that should be performed on that data.

Application of DBMS

Sector	Application
Universities	Student information, courses, grades, etc.
Sales	Customer information, sales, etc.
Finance	Stock information, sales, bonds, etc.
Banking	Customer information, account, activities, deposits, loans, etc.
Manufacturing	Production information, suppliers, inventories, etc.
Airlines	Customer information, schedules, reservations, etc.
HR Management	Employee information, payroll, deduction, paychecks, etc.
Telecommunication	Call records, bills, usage, etc.

Advantages of DBMS

- Various formats of data can be stored, and data can be retrieved by a range of methods in the Structured Query Language.
- As most databases are usually centralized in nature, they can be accessed quickly and managed easily.

- We can set up authorized users who can view, share, and access data. This ensures security for data.
- DBMS facilitates the smooth incorporation of programming languages, such as C++, Python, and PHP, to allow users to establish a connection with a web application or third-party applications.
- To prevent data from being inaccessible when there is an overload, a recovery system with automated backups is provided by almost every DBMS software.
- With minimum data duplicity and redundancy, DBMS provides data protection and integrity.

Disadvantages of DBMS

- In some cases, DBMSs are highly complicated systems to set up and maintain.
- The cost of DBMS hardware and applications is comparatively high, exhausting an organization's budget.
- In certain organizations, all information is integrated into a common database, which may get destroyed due to electrical issues or it may get corrupted in storage media. Having backups is preferred in such situations but that increases the cost.
- Certain DBMS systems cannot run complex queries as they slow down the other processes that are running.

Characteristics of DBMS

- It uses a digital repository established on a server to store and manage the information.
- It can provide a clear and logical view of the process that manipulates data.
- DBMS contains automatic backup and recovery procedures.
- It contains ACID properties which maintain data in a healthy state in case of failure.
- It can reduce the complex relationship between data.
- It is used to support manipulation and processing of data.
- It is used to provide security of data.
- It can view the database from different viewpoints according to the requirements of the user.

File system v/s Database system

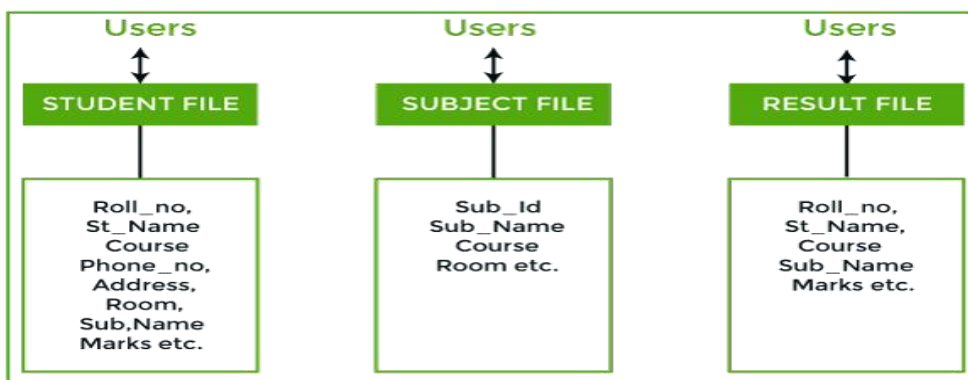
Difference between File System and DBMS:

Basis	File System	DBMS
Structure	The file system is software that manages and organizes the files in a storage medium within a computer.	DBMS is software for managing the database.
Data Redundancy	Redundant data can be present in a file system.	In DBMS there is no redundant data.
Backup and Recovery	It doesn't provide backup and recovery of data if it is lost.	It provides backup and recovery of data even if it is lost.
Query processing	There is no efficient query processing in the file system.	Efficient query processing is there in DBMS.
Consistency	There is less data consistency in the file system.	There is more data consistency because of the process of normalization.
Complexity	It is less complex as compared to DBMS.	It has more complexity in handling as compared to the file system.
Security Constraints	File systems provide less security in comparison to DBMS.	DBMS has more security mechanisms as compared to file systems.
Cost	It is less expensive than DBMS.	It has a comparatively higher cost than a file system.
Data Independence	There is no data independence.	In DBMS data independence exists.
User Access	Only one user can access data at a time.	Multiple users can access data at a time.
Meaning	The user has to write procedures for managing databases	The user not required to write procedures.
Sharing	Data is distributed in many files. So, not easy to share data	Due to centralized nature sharing is easy

Basis	File System	DBMS
Data Abstraction	It give details of storage and representation of data	It hides the internal details of Database
Integrity Constraints	Integrity Constraints are difficult to implement	Integrity constraints are easy to implement
Example	Cobol, C++	Oracle, SQL Server

File System Approach

File based systems were an early attempt to computerize the manual system. It is also called a traditional based approach in which a decentralized approach was taken where each department stored and controlled its own data with the help of a data processing specialist. The main role of a data processing specialist was to create the necessary computer file structures, and also manage the data within structures and design some application programs that create reports based on file data.



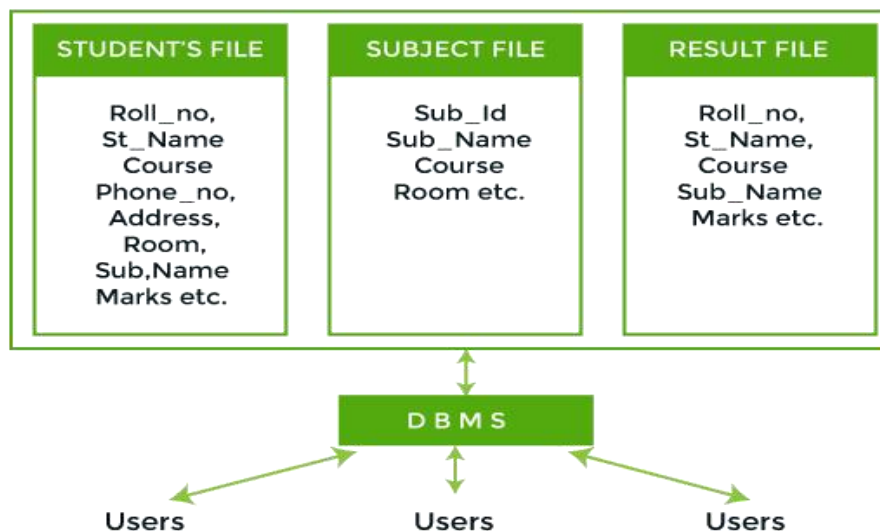
In the above figure:

Consider an example of a student's file system. The student file will contain information regarding the student (i.e. roll no, student name, course etc.). Similarly, we have a subject file that contains information about the subject and the result file which contains the information regarding the result.

Some fields are duplicated in more than one file, which leads to data redundancy. So to overcome this problem, we need to create a centralized system, i.e. DBMS approach.

DBMS:

A database approach is a well-organized collection of data that are related in a meaningful way which can be accessed by different users but stored only once in a system. The various operations performed by the DBMS system are: Insertion, deletion, selection, sorting etc.



In the above figure,

In the above figure, duplication of data is reduced due to centralization of data.

Data abstraction and data Independence

Database systems comprise complex data structures. In order to make the system efficient in terms of retrieval of data, and reduce complexity in terms of usability of users, developers use abstraction i.e. hide irrelevant details from the users. This approach simplifies database design.

There are mainly **3** levels of data abstraction:

Physical: This is the lowest level of data abstraction. It tells us how the data is actually stored in memory. The access methods like sequential or random access and file organization methods like B+ trees and hashing are used for the same. Usability, size of memory, and the number of times the records are factors that we need to know while designing the database.

Suppose we need to store the details of an employee. Blocks of storage and the amount of memory used for these purposes are kept hidden from the user.

Logical: This level comprises the information that is actually stored in the database in the form of tables. It also stores the relationship among the data entities in relatively simple structures. At this level, the information available to the user at the view level is unknown.

We can store the various attributes of an employee and relationships, e.g. with the manager can also be stored.

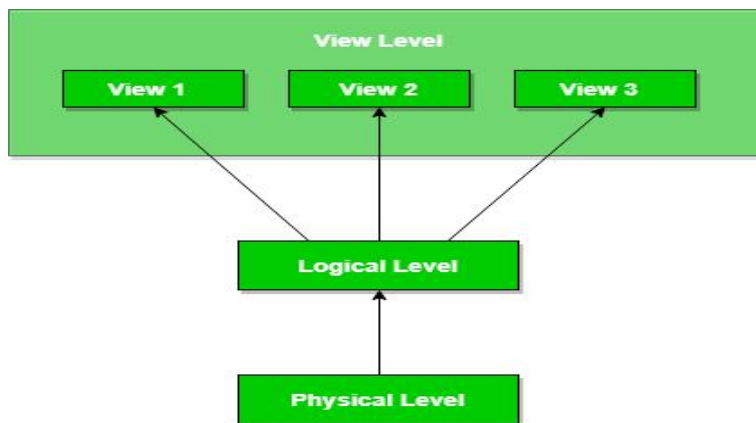
View: This is the highest level of abstraction. Only a part of the actual database is viewed by the users. This level exists to ease the accessibility of the database by an individual user. Users view data in the form of rows and columns. Tables and relations are used to store data. Multiple views of the same database may exist. Users can just view the data and interact with the database, storage and implementation details are hidden from them.

Example: In case of storing customer data,

Physical level – it will contains block of storages (bytes,GB,TB,etc)

Logical level – it will contain the fields and the attributes of data.

View level – it works with [CLI](#) or [GUI](#) access of database



The main purpose of data abstraction is to achieve data independence in order to save the time and cost required when the database is modified or altered.

Data Independence is mainly defined as a property of DBMS that helps you to change the database schema at one level of a system without requiring to change the schema at the next level. It helps to keep the data separated from all programs that make use of it.

We have namely two levels of data independence arising from these levels of abstraction :

Physical level data independence: It refers to the characteristic of being able to modify the physical schema without any alterations to the conceptual or logical schema, done for optimization purposes, e.g., the Conceptual structure of the database would not be affected by any change in storage size of the database system server. Changing from sequential to random access files is one such example. These alterations or modifications to the physical structure may include:

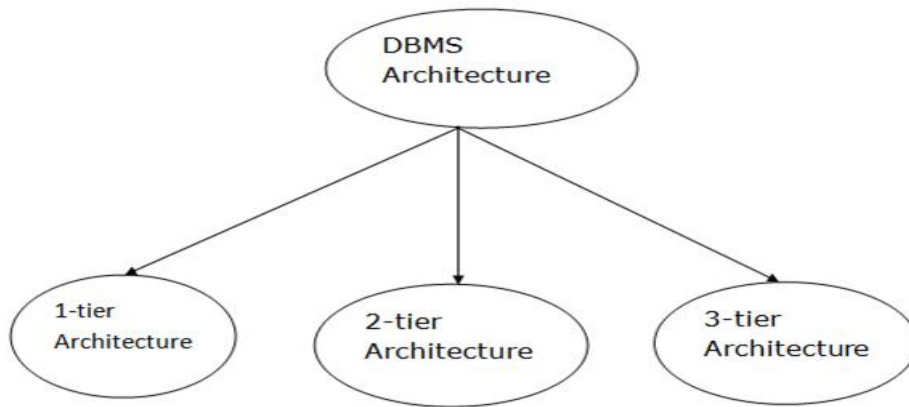
- Utilizing new storage devices.
- Modifying data structures used for storage.
- Altering indexes or using alternative file organization techniques etc.

Logical level data independence: It refers to the characteristic of being able to modify the logical schema without affecting the external schema or application program. The user view of the data would not be affected by any changes to the conceptual view of the data. These changes may include insertion or deletion of attributes, altering table structures, entities or relationships to the logical schema, etc.

DBMS Architecture

- The DBMS design depends upon its architecture. The basic client/server architecture is used to deal with a large number of PCs, web servers, database servers and other components that are connected with networks.
- The client/server architecture consists of many PCs and a workstation which are connected via the network.
- DBMS architecture depends upon how users are connected to the database to get their request done.

Types of DBMS Architecture



Database architecture can be seen as a single tier or multi-tier. But logically, database architecture is of two types like: **2-tier architecture** and **3-tier architecture**.

1-Tier Architecture

- In this architecture, the database is directly available to the user. It means the user can directly sit on the DBMS and uses it.
- Any changes done here will directly be done on the database itself. It doesn't provide a handy tool for end users.
- The 1-Tier architecture is used for development of the local application, where programmers can directly communicate with the database for the quick response.

2-Tier Architecture

- The 2-Tier architecture is same as basic client-server. In the two-tier architecture, applications on the client end can directly communicate with the database at the server side. For this interaction, API's like: **ODBC**, **JDBC** are used.
- The user interfaces and application programs are run on the client-side.
- The server side is responsible to provide the functionalities like: query processing and transaction management.
- To communicate with the DBMS, client-side application establishes a connection with the server side.

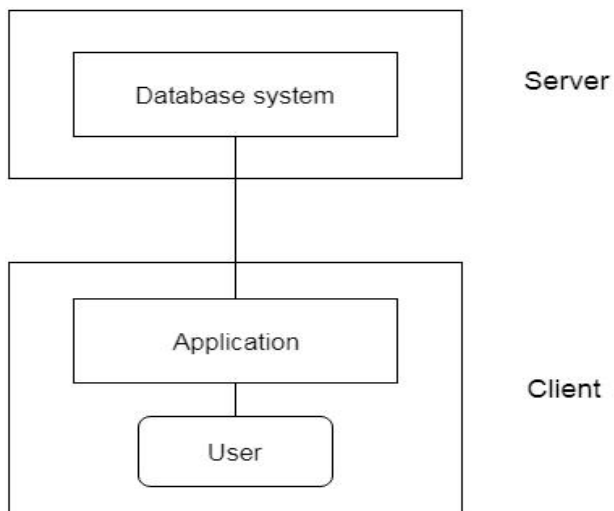


Fig: 2-tier Architecture

3-Tier Architecture

- The 3-Tier architecture contains another layer between the client and server. In this architecture, client can't directly communicate with the server.
- The application on the client-end interacts with an application server which further communicates with the database system.
- End user has no idea about the existence of the database beyond the application server. The database also has no idea about any other user beyond the application.
- The 3-Tier architecture is used in case of large web application.

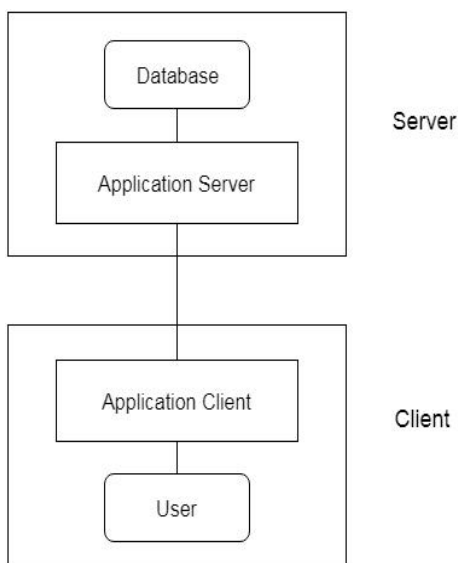
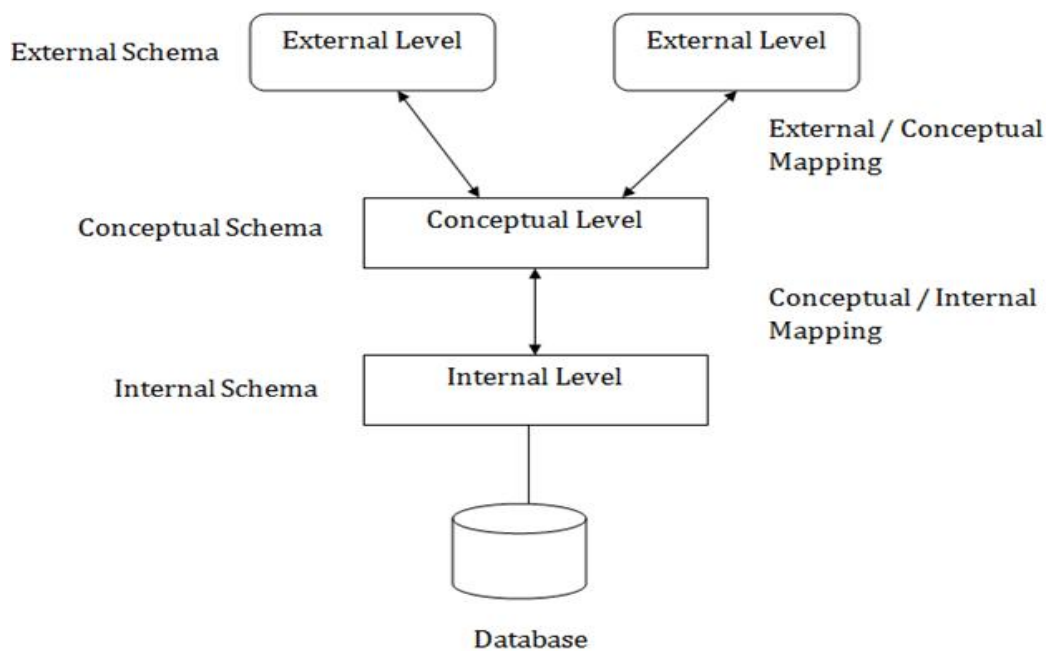


Fig: 3-tier Architecture

Three schema Architecture

- The three schema architecture is also called ANSI/SPARC architecture or three-level architecture.
- This framework is used to describe the structure of a specific database system.
- The three schema architecture is also used to separate the user applications and physical database.
- The three schema architecture contains three-levels. It breaks the database down into three different categories.

The three-schema architecture is as follows:



In the above diagram:

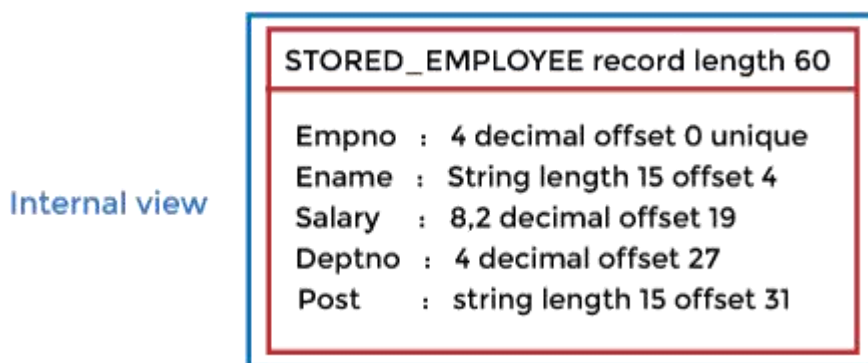
- It shows the DBMS architecture.
- Mapping is used to transform the request and response between various database levels of architecture.
- Mapping is not good for small DBMS because it takes more time.
- In External / Conceptual mapping, it is necessary to transform the request from external level to conceptual schema.
- In Conceptual / Internal mapping, DBMS transform the request from the conceptual to internal level.

Objectives of Three schema Architecture

The main objective of three level architecture is to enable multiple users to access the same data with a personalized view while storing the underlying data only once. Thus it separates the user's view from the physical structure of the database. This separation is desirable for the following reasons:

- Different users need different views of the same data.
- The approach in which a particular user needs to see the data may change over time.
- The users of the database should not worry about the physical implementation and internal workings of the database such as data compression and encryption techniques, hashing, optimization of the internal structures etc.
- All users should be able to access the same data according to their requirements.
- DBA should be able to change the conceptual structure of the database without affecting the user's
- Internal structure of the database should be unaffected by changes to physical aspects of the storage.

1. Internal Level



- The internal level has an internal schema which describes the physical storage structure of the database.
- The internal schema is also known as a physical schema.
- It uses the physical data model. It is used to define that how the data will be stored in a block.
- The physical level is used to describe complex low-level data structures in detail.

The internal level is generally is concerned with the following activities:

- Storage space allocations.

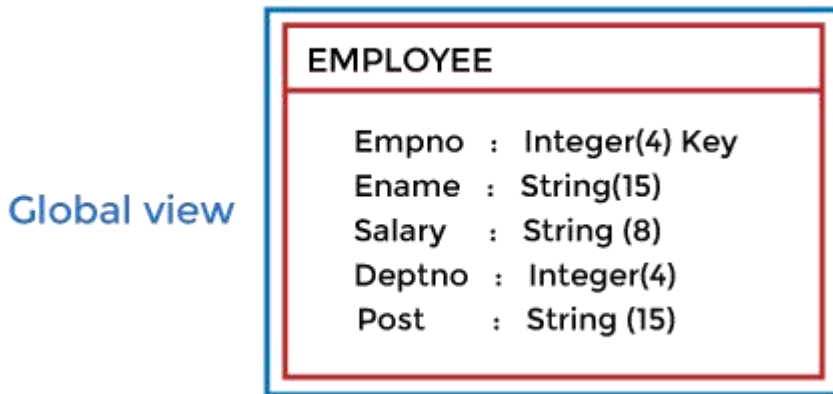
For Example: B-Trees, Hashing etc.

- Access paths.

For Example: Specification of primary and secondary keys, indexes, pointers and sequencing.

- Data compression and encryption techniques.
- Optimization of internal structures.
- Representation of stored fields.

2. Conceptual Level



- The conceptual schema describes the design of a database at the conceptual level. Conceptual level is also known as logical level.
- The conceptual schema describes the structure of the whole database.
- The conceptual level describes what data are to be stored in the database and also describes what relationship exists among those data.
- In the conceptual level, internal details such as an implementation of the data structure are hidden.
- Programmers and database administrators work at this level.

3. External Level



- At the external level, a database contains several schemas that sometimes called as subschema. The subschema is used to describe the different view of the database.
- An external schema is also known as view schema.
- Each view schema describes the database part that a particular user group is interested and hides the remaining database from that user group.
- The view schema describes the end user interaction with database systems.

Describe overall architecture of DBMS with diagram.

The architecture of a database system is greatly influenced by the underlying computer system on which the database is running:

- i. Centralized.
- ii. Client-server.
- iii. Parallel (multi-processor).
- iv. Distributed

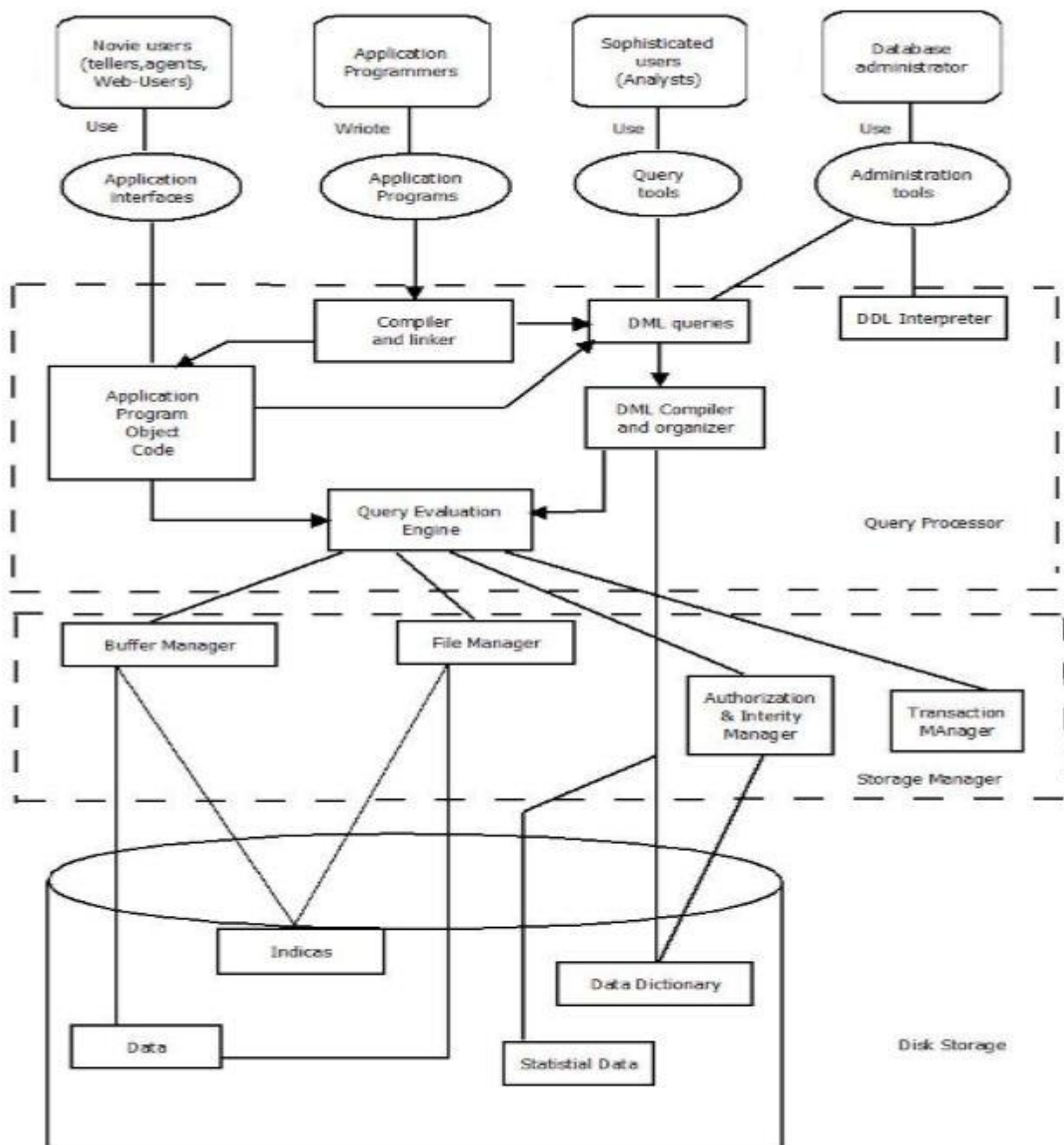


Fig1. Database System Architecture.

Database Users:

Users are differentiated by the way they expect to interact with the system:

Application programmers:

- Application programmers are computer professionals who write application programs. Application programmers can choose from many tools to develop user interfaces.
- Rapid application development (RAD) tools are tools that enable an application programmer to construct forms and reports without writing a program.

Sophisticated users:

- Sophisticated users interact with the system without writing programs. Instead, they form their requests in a database query language.
- They submit each such query to a query processor, whose function is to break down DML statements into instructions that the storage manager understands.

Specialized users :

- Specialized users are sophisticated users who write specialized database applications that do not fit into the traditional data-processing framework.
- Among these applications are computer-aided design systems, knowledge base and expert systems, systems that store data with complex data types (for example, graphics data and audio data), and environment-modeling systems.

Naïve users :

- Naive users are unsophisticated users who interact with the system by invoking one of the application programs that have been written previously.
- For example, a bank teller who needs to transfer \$50 from account A to account B invokes a program called transfer. This program asks the teller for the amount of money to be transferred, the account from which the money is to be transferred, and the account to which the money is to be transferred.

Database Administrator:

- Coordinates all the activities of the database system. The database administrator has a good understanding of the enterprise's information resources and needs.
- Database administrator's duties include:
 - **Schema definition:** The DBA creates the original database schema by executing a set of data definition statements in the DDL.
 - **Storage structure and access method definition.**

- **Schema and physical organization modification:** The DBA carries out changes to the schema and physical organization to reflect the changing needs of the organization, or to alter the physical organization to improve performance.
- **Granting user authority to access the database:** By granting different types of authorization, the database administrator can regulate which parts of the database various users can access.
- **Specifying integrity constraints.**
- **Monitoring performance and responding to changes in requirements.**

Query Processor:

The query processor will accept query from user and solves it by accessing the database.

Parts of Query processor:

DDL interpreter

This will interprets DDL statements and fetch the definitions in the data dictionary.

DML compiler

- a. This will translates DML statements in a query language into low level instructions that the query evaluation engine understands.
- b. A query can usually be translated into any of a number of alternative evaluation plans for same query result DML compiler will select best plan for query optimization.

Query evaluation engine

This engine will execute low-level instructions generated by the DML compiler on DBMS.

Storage Manager/Storage Management:

- A storage manager is a program module which acts like interface between the data stored in a database and the application programs and queries submitted to the system.
- Thus, the storage manager is responsible for storing, retrieving and updating data in the database.
- **The storage manager components include:**
 - **Authorization and integrity manager:** Checks for integrity constraints and authority of users to access data.
 - **Transaction manager:** Ensures that the database remains in a consistent state although there are system failures.
 - **File manager:** Manages the allocation of space on disk storage and the data structures used to represent information stored on disk.

- **Buffer manager:** It is responsible for retrieving data from disk storage into main memory. It enables the database to handle data sizes that are much larger than the size of main memory.
- **Data structures implemented by storage manager.**
- **Data files:** Stored in the database itself.
- **Data dictionary:** Stores metadata about the structure of the database.
- **Indices:** Provide fast access to data items.

Database Administrator

A **Database Administrator (DBA)** is an individual or person responsible for controlling, maintaining, coordinating, and operating a database management system. Managing, securing, and taking care of the database systems is a prime responsibility. They are responsible and in charge of authorizing access to the database, coordinating, capacity, planning, installation, and monitoring uses, and acquiring and gathering software and hardware resources as and when needed. Their role also varies from configuration, database design, migration, security, troubleshooting, backup, and data recovery. Database administration is a major and key function in any firm or organization that is relying on one or more databases. They are overall commanders of the Database system.

Importance of Database Administrator (DBA) :

- Database Administrator manages and controls three levels of database internal level, conceptual level, and external level of Database management system architecture and in discussion with the comprehensive user community, gives a definition of the world view of the database. It then provides an external view of different users and applications.
- Database Administrator ensures held responsible to maintain integrity and security of database restricting from unauthorized users. It grants permission to users of the database and contains a profile of each and every user in the database.
- Database Administrators are also held accountable that the database is protected and secured and that any chance of data loss keeps at a minimum.
- Database Administrator is solely responsible for reducing the risk of data loss as it backup the data at regular intervals.

Role and Duties of Database Administrator (DBA) :

- **Decides hardware –**
They decide on economical hardware, based on cost, performance, and efficiency of hardware, and best suits the organization. It is hardware that is an interface between end users and the database.
- **Manages data integrity and security –**
Data integrity needs to be checked and managed accurately as it protects and restricts data from unauthorized use. DBA eyes on relationships within data to maintain data integrity.
- **Database Accessibility –**
Database Administrator is solely responsible for giving permission to access data available in the database. It also makes sure who has the right to change the content.
- **Database design –**
DBA is held responsible and accountable for logical, physical design, external model design, and integrity and security control.

- **Database implementation –**

DBA implements DBMS and checks database loading at the time of its implementation.

- **Query processing performance –**

DBA enhances query processing by improving speed, performance, and accuracy.

- **Tuning Database Performance –**

If the user is not able to get data speedily and accurately then it may lose organization's business.

So by tuning SQL commands DBA can enhance the performance of the database.

Various responsibilities of Database Administrator (DBA) :

- Responsible for designing overall database schema (tables & fields).
- To select and install database software and hardware.
- Responsible for deciding on access methods and data storage.
- DBA selects appropriate DBMS software like oracle, SQL server or MySQL.
- Used in designing recovery procedures.
- DBA decides the user access level and security checks for accessing, modifying or manipulating data.
- DBA is responsible for specifying various techniques for monitoring the database performance.
- DBA is responsible for operation managements.
- The operation management deals with the data problems which arises on day to day basis, and the responsibilities include are:
 1. Investigating if any error is been found in the data.
 2. Supervising of restart and recovery procedures in case of any event failure.
 3. Supervising reorganization of the databases.
 4. Controlling and handling all periodic dumps of data.